Catalog observations and limits on reproducing earthquake-triggered activity at Sierra Negra

Reproducibility project

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Abstract

Earthquake timing and waveform recurrence can constrain the processes connecting fault slip to eruption. We assess the reproducibility of Song et al. (2026) and calculate seismicity statistics from the independently archived original catalog. The full archived catalog passes its published checksum and contains 219 earthquakes on 26 June 2018. Their distribution among four published activity windows is 29, 74, 72, and 44 events. We implement and test spectral classification, signed cross-correlation, connected-component clustering, magnitude-to-moment conversion, and recurrence statistics. We acquire nine three-component seismic station records, eight complete and one gapped, and execute a public picker on the seven 8G stations producing 6,908 P and 6,379 S candidates, review the supplementary methods, and independently process four historical PAYG day plots. Missing numerical velocity layers, station records and source controls prevent a faithful association and relocation reproduction. Therefore, no enhanced catalog, long-period classifications, multiplet memberships, or published recurrence regressions are independently reproduced. Partial observation-based figures and an original conceptual schematic document the available calculations. The results establish a reusable numerical basis and a transparent assessment of missing inputs; they do not independently verify a fluid-triggering mechanism.

1 Introduction

Volcanic eruption initiation involves mechanical interaction between a magma reservoir and the surrounding faulted edifice. Seismic timing alone leaves an ambiguity: fault slip may open a magma pathway directly, or it may initiate a sequence of processes that eventually permits intrusion. The Sierra Negra observations motivate testing this distinction because a well-recorded fault earthquake preceded the 2018 eruption (Bell et al., 2021; Song et al., 2026).

An enhanced earthquake catalog can reveal signals absent from an original monitoring catalog, but agreement depends on waveform coverage, detection thresholds, and manual decisions. We distinguish three reproducibility questions: whether an archived original catalog can be recovered; whether numerical components behave as defined; and whether the enhanced observations and mechanism can be independently verified. Only the first two questions are answered by the present calculations. The project assessment explicitly retains the third as unresolved.

2 Data availability and provenance

The original earthquake catalog is deposited with geodetic and interferometric products by Bell et al. (2020). We downloaded the complete event list and verified its published MD5. The date-specific subset confirms all 219 previously recovered browser rows. It retains identifiers, origin

times, coordinates, catalog depths, magnitudes, location-error fields, travel-time RMS, azimuthal gap, and phase-pick counts. Full-source SHA-256 and archive provenance accompany the derived catalog. Zenodo metadata specify CC-BY-4.0 licensing. The archived long-term GNSS baseline also passes its published checksum but does not replace high-rate positions.

Public metadata identify IGUANA as network 8G during 2018–2019 (Bell et al., 2018). Network access permits downloading 21 complete 100-Hz channels from seven 8G stations: SN04, SN05, SN07, SN11, SN12, SN13 and SN14. Two additional EC station records are acquired with responses: SN06 is complete, while SN02 has 79,636.67 seconds of common three-component coverage (92.172%). A consolidated map displays fourteen seismic metadata stations, distinguishing the nine acquired stations from metadata-only entries. Missing records remain a coverage limitation. The public Volpick repository provides trained models (Zhong, 2026; Zhong and Tan, 2024), and public checkpoint v1 is downloaded on the data drive. Its equivalence to the authors' checkpoint is unverified. The separate VCH1 original catalog passes its published checksum and contains 191 entries on the date; it is not merged with the located catalog because event equivalence is unverified. The archived scalar GNSS baseline is acquired, but no high-rate position solutions are obtained. The full archived VCH1 catalog also matches its published checksum and contributes 191 entries on the date. These time/magnitude entries remain separate from the 219 located events because shared event identities and locations are unavailable for reliable merging. GV03 and GV09 daily-file requests return provider login HTML. Station coordinates from DOI pages do not constitute deformation observations.

The accessible main article does not expose a separate Code Availability section or an enhanced catalog download. Targeted searches did not identify a study-specific code deposit. This does not establish that such a deposit is absent. The supplementary PDF is downloaded and its methods, equations and nine figures are reviewed. Picking uses 100-Hz inputs filtered at 2–12 Hz, 60-second windows with 50% overlap and 0.2 phase thresholds. Association requires four P and one S arrival with a 0.5-second residual tolerance. The cited velocity-model manuscript (Rezaeifar et al., 2024) displays a P model but does not provide numerical layer values identified in the accessible text. No layers or S-velocity prescription are invented.

3 Methods

3.1 Original-catalog calculations

We parse source rows without rejecting events by location uncertainty, magnitude, or depth. Validation requires unique identifiers, finite numerical fields, valid coordinate ranges, and chronological ordering. UTC boundaries of 09:15, 17:00, and 19:40 define half-open phase intervals. An event exactly on a boundary belongs to the later phase. This uses the publication's rounded times; the largest catalog event occurs 32.299 seconds after the first boundary and is consequently counted in Phase 2.

For phase k, the observed catalog rate is $r_k = N_k/\Delta t_k$, where N_k is the number of catalog entries and Δt_k is the interval duration in hours. This rate describes the catalog. It is not corrected for completeness, missing stations, or time-dependent noise. Maps preserve source coordinates; zoomed panels state how many events lie outside their extent. The latitude–depth plot includes every event rather than imposing an undocumented cross-section selection. The source depth datum is not verified, so the plotted values are labeled catalog depth.

3.2 Waveform and classification components

Waveform processing requires contiguous finite segments. The implementation rejects gaps instead of silently filling them. Detrending, Butterworth bandpass filtering, and rational anti-alias resampling are available, but filter order and phase convention must be specified. Instrument-response removal is a dependency-bound ObsPy operation with an explicit stabilization passband. None of these operations were applied to Sierra Negra waveforms in this run.

The classification kernel calculates

$$F = \log_2 \left(\frac{\overline{A}_{9\text{–}15}}{\overline{A}_{2\text{–}5}} \right), \tag{1}$$

where $\overline{A}$ is mean Fourier amplitude in the indicated frequency band, in hertz. Five-second arrival windows can be extracted from one second before a pick. This expression measures amplitude contrast; it is not an integrated power ratio. Station estimates are averaged only when their supplied signal-to-noise ratios satisfy the criterion of two. Values below -2 are LP, and values at or above -2 are VT. Missing measurements remain unclassified. These definitions follow the accessible method description (Song et al., 2026); taper, SNR definition, and filter details remain unresolved.

3.3 Correlation and families

Continuous template correlation uses FFT convolution through compiled SciPy routines. Moving-window sums and sums of squares give the local normalization. Constant templates and non-finite samples are rejected; zero-energy data windows return undefined scores. Detection requires explicit thresholds and separation. We implement both absolute and the EQcorrscan median-absolute-score convention (EQcorrscan contributors, 2026); a chosen sensitivity threshold would not be labeled a published threshold.

An independent C++ kernel demeans each overlapping event-pair window, computes normalized cross-correlation at a user-specified range of integer lags, and retains the sign of the coefficient with largest absolute value. Thus a polarity-reversed signal remains negatively correlated. Common station/component/phase windows contribute to a network average; missing channels contribute no artificial zero. This equal-channel weighting and lag selection are independent conventions awaiting comparison with supplementary details.

Positive network coefficients greater than 0.7 create graph edges. Connected components with at least ten events define candidate families; explicit template links can propagate family and FI membership. Negative coefficients below -0.7 create review candidates, not automatically confirmed anti-repeaters. Observed polarity reversals require station-by-station inspection. No observational similarity graph was constructed.

3.4 Recurrence, moment, and geodesy

For chronologically ordered family events, recurrence intervals are $T_i = t_{i+1} - t_i$. Periodicity is their mean divided by their standard deviation. The implementation exposes the population versus sample standard-deviation choice because that convention is not resolved. Duplicated origin times require review.

The accessible moment conversion is

$$M_0 = 10^{1.5M_w + 9.105}\ \mathrm{N\,m}, \qquad M_w = \begin{cases} M_L, & M_L > 3, \\ 2M_L/3 + 1, & M_L \le 3. \end{cases} \tag{2}$$

Its application to a new catalog requires confirming the magnitude scale. The regression functions require explicit linear or logarithmic coordinates and an explicit choice of pairing moment with the preceding or following recurrence interval. Leave-one-out jackknife calculations estimate slope variability. These choices cannot be selected by optimizing agreement with a target coefficient.

The GNSS utility differences average positions in one-hour windows before and after an earthquake. It does not estimate precise point positions. A full geodetic reproduction requires observations, correction products, software, and control settings. Likewise, phase association and differential relocation require verified velocity models and input controls (Münchmeyer, 2024; Waldhauser, 2001); neither was executed.

4 Results

The recovered original catalog contains 219 earthquakes, with a magnitude range from -0.63 to 4.73 and median 1.78. The largest entry, dias2018mkgw, has an origin time of 09:15:32.299 UTC. Its catalog magnitude is retained as 4.73; it is not substituted for the separately reported moment magnitude of the fault earthquake.

Phase	Duration (h)	Events	Rate (h^{-1})	Median magnitude
1	9.250	29	3.135	0.900
2	7.750	74	9.548	0.845
3	2.667	72	27.000	2.935
4	4.333	44	10.154	2.520

Table 1: Calculations from the original archived catalog. UTC boundaries are 00:00, 09:15, 17:00, 19:40, and 24:00. Counts include all source events and are not completeness-corrected.

Figure 1 shows the spatial observations recoverable without waveform processing. One location lies far from the caldera vicinity; preserving it makes the distinction between source data and an interpreted local population explicit. No new location inversion or structural reconstruction is implied. Figure 2 and Table 1 show the highest observed catalog rate during the third interval. The corresponding rise in median magnitude means that changing detection completeness must be considered before comparing physical event rates. Using the alternative published eruption boundary of 19:35 changes Phase 3 and 4 counts to 69 and 47, respectively. Three events therefore change phase assignment within the five-minute timing range.

Eleven numerical test cases pass. FFT correlation agrees with direct Pearson correlation, and the C++ signed lag kernel agrees with a NumPy reference. Injected positive and negative signals are recovered at their known positions. A constructed Fourier spectrum verifies the amplitude-based FI boundary. Additional tests check transitive components, magnitude conversion, recurrence pairing, regression and jackknife calculations, arrival-window boundaries, and geodetic differencing. A synthetic end-to-end window-CLI test verifies classification, family membership and interval output. A further overlap test verifies probability averaging, blind edges and refusal of incomplete three-component inputs. Constructed test signals are algorithm validation data and are excluded from observational products.

4.1 Candidate phase-picking execution

We execute the public Volpick checkpoint on all 168 acquired 8G station-hours, obtaining 6,908 candidate P picks and 6,379 candidate S picks. The supplement reports 13,144 P and 12,196 S

candidates before association across its study network; our differences are therefore $-6,236$ and $-5,817$, respectively. Seven-station coverage and independently specified processing/checkpoint details prevent interpreting these differences as event completeness. The published 6,261 P and 5,159 S main-text arrivals instead refer to associated arrivals, so they are a different comparison target.

The asynchronous SeisBench stream wrapper stalled in this environment. Direct execution uses the checkpoint's own batch preprocessing, forward and postprocessing methods, with averaged finite overlapping probabilities. We use 60-second windows at 50% overlap, a four-corner causal 2–12 Hz filter, per-window pretrained-model normalization, five-second edge blinding and the package's probability-trigger routine at 0.2. Filter order/phase, normalization, blinding, stacking and exact checkpoint equivalence are not established as the original implementation. Checkpoint hashes, model settings and per-hour execution records are saved. No false-pick rate, event completeness or new hypocenter is inferred from these candidate arrivals.

4.2 Independent reproduction of PAYG day plots

We obtain four full-day IU.PAYG.00.BHZ records corresponding to Supplementary Figure 5. We remove the epoch instrument response to recover velocity in m/s, using a 0.01, 0.02, 8, 9 Hz stabilization prefilter and no water level. These are independent processing choices because the original stabilization settings are unspecified. Each contiguous segment is processed separately; gaps are not interpolated. Thirty-minute rows use one-second minimum/maximum envelopes. Per-day display gain is the 99th percentile of absolute velocity, with no amplitude clipping. Major wave packets occur in the corresponding published time rows, but identical amplitudes cannot be claimed without matching preprocessing and gain.

5 Comparison with published results

The comparison table in the repository distinguishes uncomputed quantities from numerical disagreements. The accessible paper reports two inconsistent VT counts: 1,224 in Results and 1,214 in Methods. With 711 LP events, the first total is 1,935 and the second is 1,925. Independently, 694 located events plus 1,241 new template detections sum to 1,935. We preserve both reported counts and request source catalog reconciliation before selecting a target.

Quantity	Published	Present calculation
Enhanced event total	1,935 by stated workflow	Not computed
VT classifications	1,224 / 1,214	Not computed
LP classifications	711	Not computed
Families / member events	29 / 714	Not computed
Anti-correlated LP pairs	11	Not computed
VT / LP regression R^2	0.70 / 0.07	Not computed
Fault earthquake time	09:15 UTC	Original entry 09:15:32.299
Original catalog subset	Archive rows for date	219 events recovered

Table 2: Comparison targets from Song et al. (2026) and computations from the original archive. The 219 original events are not a numerical estimate of enhanced-catalog completeness. No difference is assigned to uncomputed quantities.

Exact agreement of the enhanced waveform results is not testable here. Main Figures 3 and

5 require FI measurements, family memberships, and original waveform examples. The reviewed supplement contains S1–S9. We reproduce all four PAYG ground-velocity day plots in S5 from archived waveforms and epoch responses (Figure 6); other supplementary figures remain unavailable. S9 fits log–log moment and recurrence time using RANSAC. Its unspecified estimator controls prevent exact regression agreement; the existing ordinary-least-squares routine is an independent numerical option.

6 Discussion

Figure 7 is an original schematic separating recorded seismic observations from the proposed physical sequence. A delayed response could be consistent with local changes in fault conditions, but the original catalog alone does not measure fluid pressure, flux, or effective stress. The absence of reproduced waveform and geodetic constraints prevents an independent test of the mechanism.

A rigorous recurrence analysis must also consider missed events, temporal censoring, magnitude uncertainty, and dependence among events in the same family. These effects can alter a moment–recurrence regression even when pointwise numerical methods are correct. Leave-one-out resampling of individual events does not account for all family-level dependence. Future observational execution should compare both interval-pairing conventions, quantify family-level uncertainty, and inspect how frequency bands and station weights alter classifications and membership.

7 Limitations

Network access permits archive downloads, although some endpoints still fail. The modern Python environment includes ObsPy, PyTorch, SeisBench and PyOcto. The public GitHub repository is created and its pushed commit verified at the public project repository. VCH1 waveform access, SN02 gaps, numerical velocity layers and source controls limit an exact enhanced-catalog reproduction. Newly acquired EC data have not been picked or associated. EarthScope GNSS file requests redirect to a separate provider login; the legacy station download links redirect to the provider home page. No GNSS raw-file existence or successful acquisition is claimed from these redirects.

The supplement has been reviewed and archive checksums and license are verified. Source depth datum and location-error units remain unresolved. No enhanced catalog, event-associated LP/VT population, observed family membership, polarity reversal, high-rate GNSS solution or published regression is claimed. Independent response stabilization and graphical gain choices limit amplitude comparison for the reproduced PAYG figure.

8 Conclusions

We recover a complete date-specific subset of the archived original catalog, calculate its phase statistics, and verify numerical components needed for further reproduction. The central completed observational result is a 219-event original catalog with a peak recorded rate of 27 events per hour in the published intrusion interval. This does not independently establish the LP population or the proposed triggering mechanism. Complete station coverage, numerical velocity layers, original family data and geodetic solutions are required to test those findings. The acquired waveforms and reviewed methods support further processing, subject to missing numerical velocity layers, source controls and station coverage.

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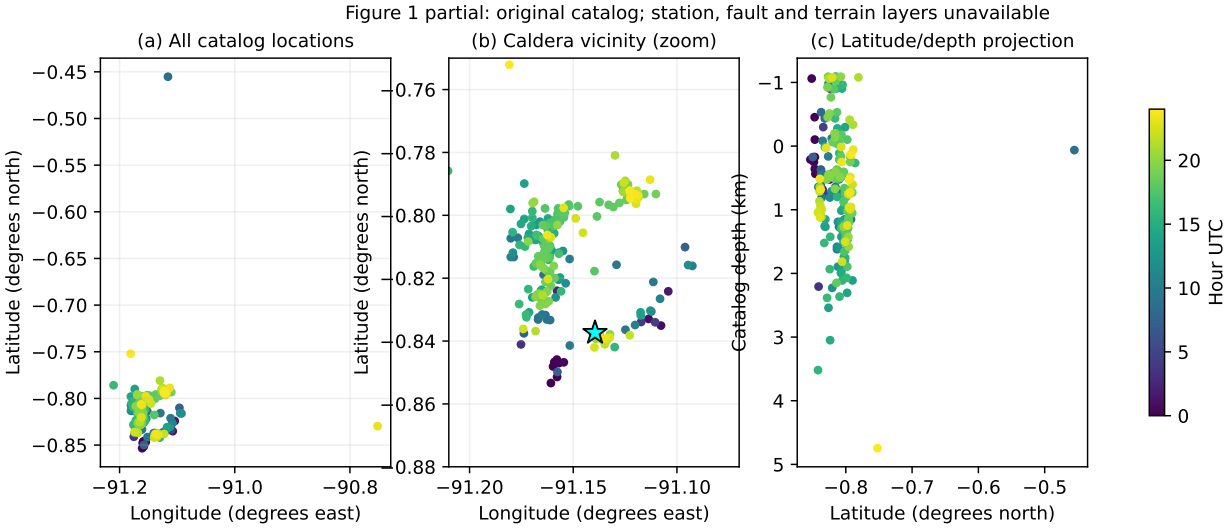


Figure 1: Partial reproduction corresponding to main Figure 1. Original Bell archive event coordinates are colored by UTC hour. Panel (a) preserves every catalog location; (b) zooms into the caldera vicinity and marks the maximum-magnitude entry; (c) projects all events onto latitude and source catalog depth. Station layers are supplied separately; this panel lacks fault, terrain and reservoir layers, and the depth datum is unresolved. These are observations from the original catalog, not an enhanced-catalog reconstruction.

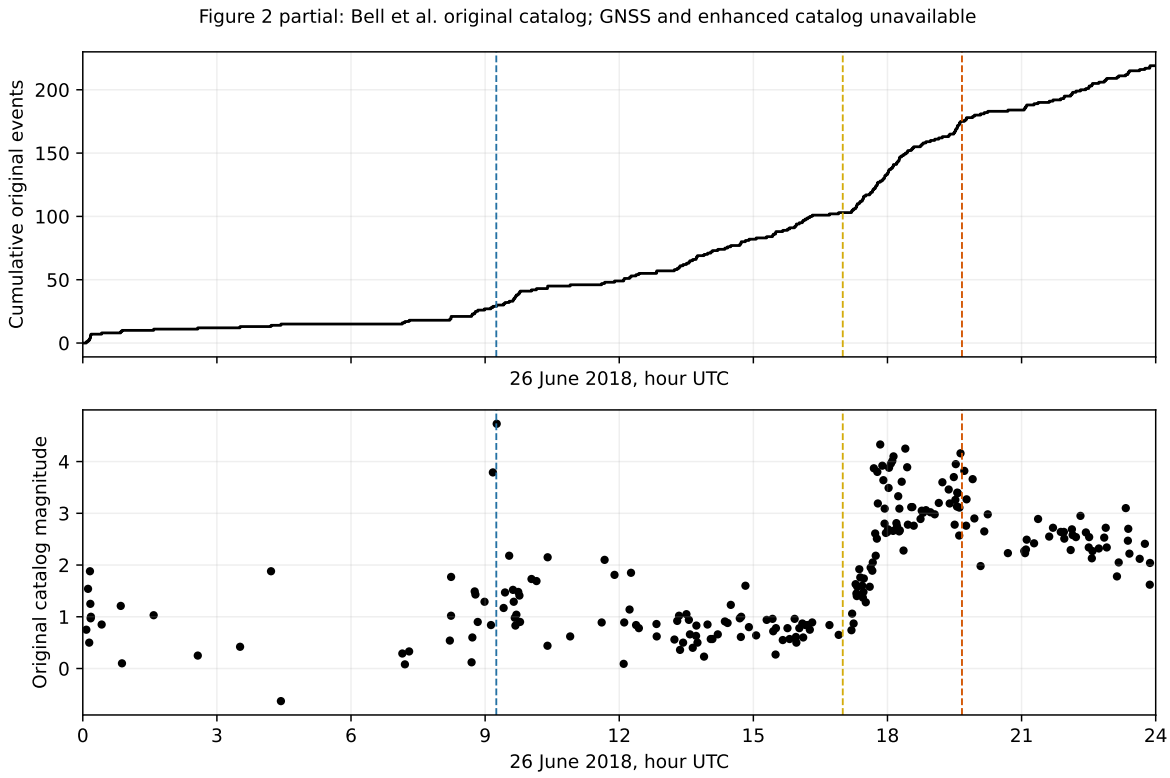


Figure 2: Partial reproduction corresponding to main Figure 2. Cumulative original events and original catalog magnitudes are plotted against hour UTC on 26 June 2018. Dashed lines give published rounded fault-earthquake, intrusion, and eruption times. Enhanced counts and GNSS displacement are absent. The timing lines are literature annotations rather than measurements newly extracted from waveforms or deformation.

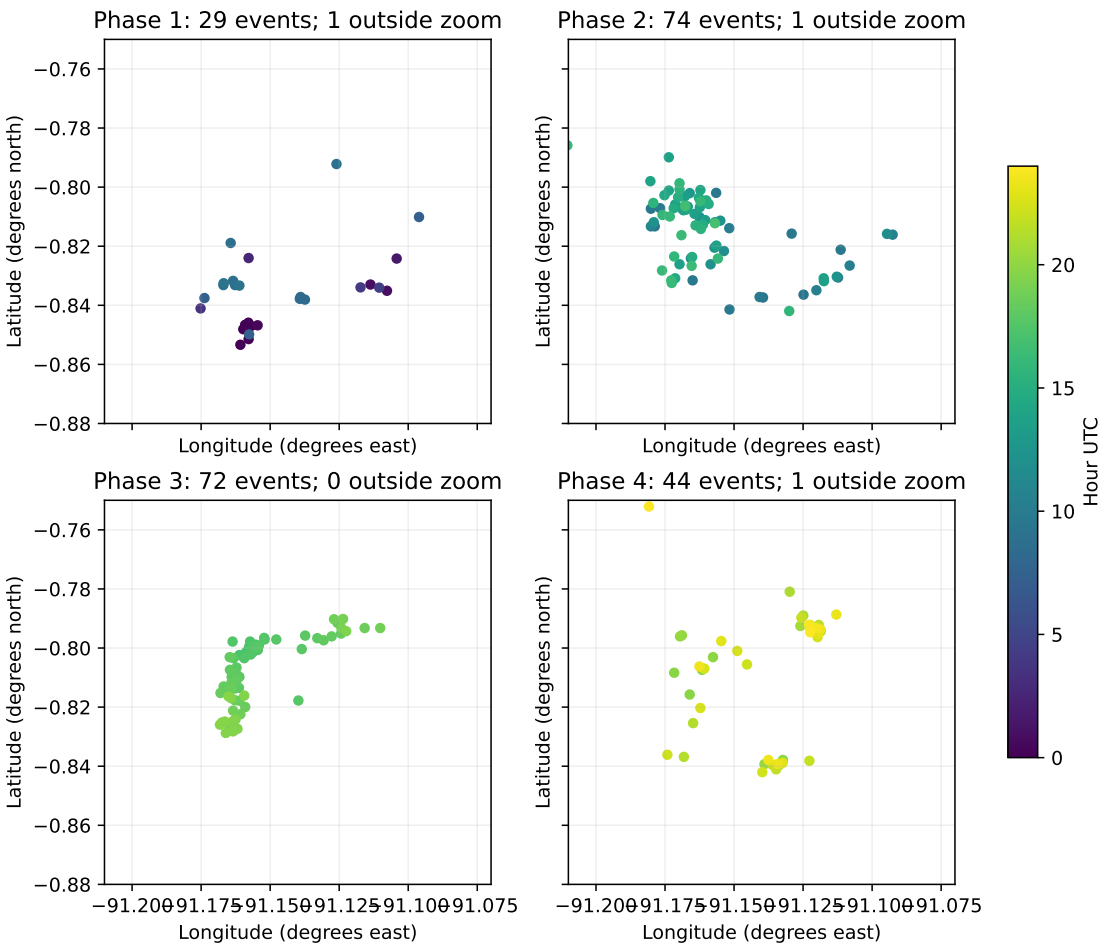


Figure 3: Partial reproduction corresponding to main Figure 4. Each panel plots original event locations in one published phase, with color indicating UTC hour. Titles report full phase counts and locations outside the zoom. Frequency-index color, topography, fault traces, and GNSS records remain unavailable. This comparison documents temporal changes in the original catalog without assigning unmeasured LP or VT labels.

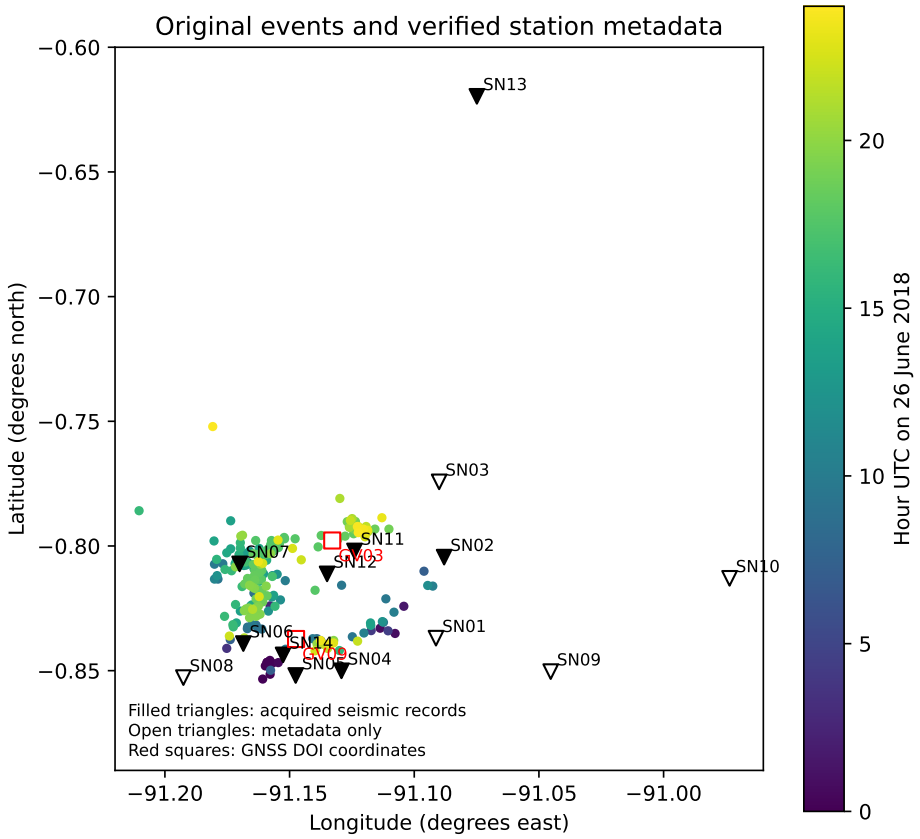


Figure 4: Additional reconstruction of station distribution for main Figure 1. Triangles show fourteen archived 8G/EC seismic metadata stations. Filled triangles mark the nine stations with acquired waveforms; open triangles mark metadata-only stations. Red squares give rounded GV03/GV09 coordinates from DOI landing pages; their coordinate datum is not independently verified. Original-catalog events are colored by UTC hour. Terrain and faults remain absent. Metadata presence is not evidence of recording availability.

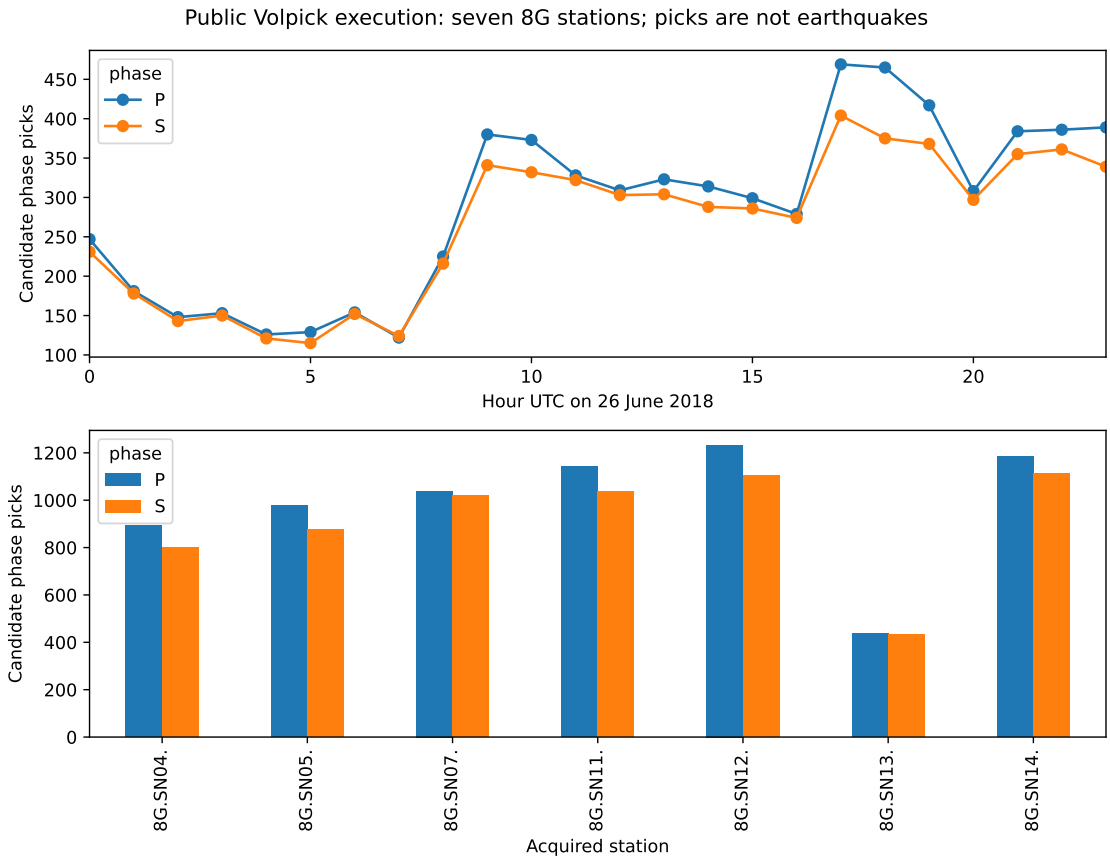


Figure 5: Candidate phase picks from the completed public-model execution on seven acquired 8G stations. Upper panel counts candidates by UTC hour; lower panel counts them by station. P and S denote model-assigned phase labels, not independently validated event arrivals. Differences in station coverage, noise and model response affect the distributions. This diagnostic is not a reproduction of the enhanced earthquake count curve.

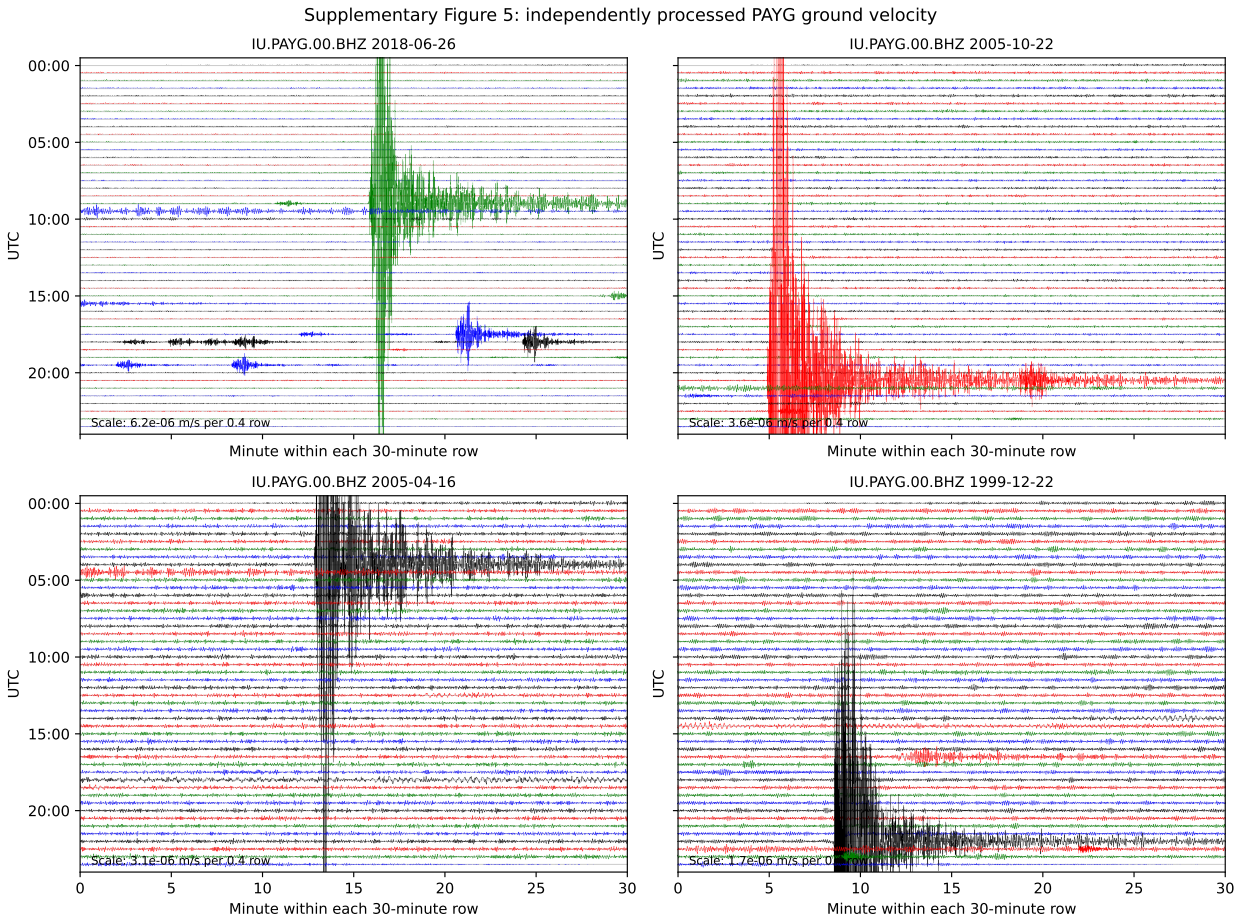


Figure 6: Independent computational reproduction of Supplementary Figure 5 from raw PAYG vertical records and instrument responses. Panels cover 26 June 2018, 22 October 2005, 16 April 2005 and 22 December 1999. All four records span a complete day. Color cycles across thirty-minute rows; ordinate labels indicate UTC and the abscissa gives minutes within each row. Insets state the separate velocity display scales. Response stabilization and display gain are independent documented choices, so this comparison tests timing and morphology rather than exact amplitude agreement.

Figure 6: original conceptual interpretation; no numerical simulation; geometry not to scale

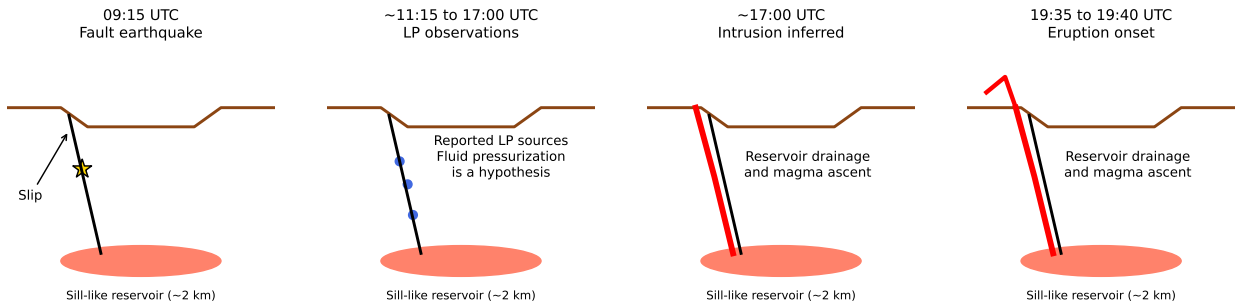


Figure 7: Original conceptual interpretation corresponding to main Figure 6. The panels depict fault slip, a proposed intervening fluid process associated with reported LP sources, intrusion, and eruption. Times are taken from the literature, including its 19:35–19:40 onset range. Geometry is not to scale. Fluid pressurization is a hypothesis; this is neither a numerical simulation nor an independently measured pressure history.